



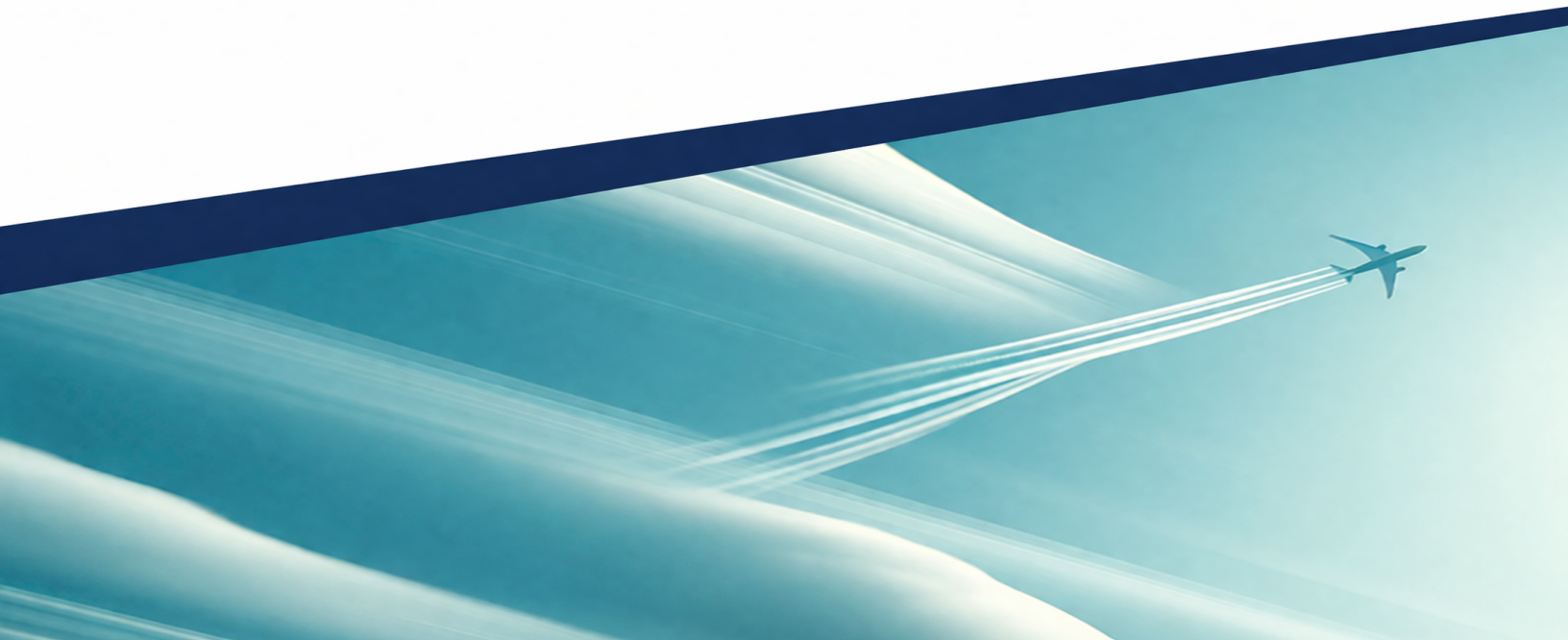
POLITECNICO
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ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

POLITECNICO DI MILANO
Laurea Magistrale – Aeronautical Engineering

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Problem Setup

According to the datas, the wing is free to bend and twist, with a strut attached at $y = y_B$. The aerodynamic center (AC) is located at e from the elastic axis. The aileron is located at the wing tip, with span b_a and deflection β .

Thus, the wing has two degrees of freedom (DOFs) :

- $w(y, t)$: Bending deformation along $\mathbf{z}$
- $\theta(y, t)$: Rotation around $\mathbf{y}$

The goal is to develop a numerical model for wing bending and torsion based on the Ritz–Galerkin approach to perform a static aeroelastic analysis.

Divergence Dynamic Pressure q_D - Derivation

Shape Functions

The wing is clamped at the root. Essential boundary conditions :

For w : $w(0) = 0$ (No displacement), $w'(0) = 0$ (No bending)

For θ : $\theta(0) = 0$ (No rotation)

We approximate the problem with the Ritz–Galerkin approach :

$$w(y, t) = N_w(y) \phi_w(t), \quad \theta(y, t) = N_\theta(y) \phi_\theta(t) \quad (1)$$

To comply with the essential boundary conditions, the shape functions must satisfy :

— N_w : at least 2nd order (2 BCs to satisfy)

— N_θ : at least 1st order (1 BC to satisfy)

For N_w , we take a parabolic shape function :

$$N_w(y) = A_w \left(\frac{y}{b}\right)^2 + B_w \left(\frac{y}{b}\right) + C_w$$

Applying BCs : $N_w(0) = 0 \Rightarrow C_w = 0$, $N'_w(0) = 0 \Rightarrow B_w = 0$; Normalizing : $N_w(b) = 1 \Rightarrow A_w = 1$.

$$\boxed{N_w(y) = \left(\frac{y}{b}\right)^2 \Rightarrow w(y, t) = \left(\frac{y}{b}\right)^2 \phi_w(t)} \quad (2)$$

For N_θ , we take an affine shape function :

$$N_\theta(y) = A_\theta \frac{y}{b} + B_\theta$$

Applying BC : $N_\theta(0) = 0 \Rightarrow B_\theta = 0$; Normalizing $N_\theta(b) = 1 \Rightarrow A_\theta = 1$.

$$\boxed{N_\theta(y) = \frac{y}{b} \Rightarrow \theta(y, t) = \frac{y}{b} \phi_\theta(t)} \quad (3)$$

Principle of Virtual Work

In order to compute q_D , we need to find the equations of motion. To do so, we use the PVW : We need to compute the internal and external virtual work.

Internal Virtual Work

$$\delta W_i = \int_0^b \delta w_{,yy} EJ w_{,yy} dy + \int_0^b \delta \theta_{,y} GJ \theta_{,y} dy \quad (4)$$

$$= \int_0^b \delta \phi_w \frac{2}{b^2} EJ \frac{2}{b^2} \phi_w dy + \int_0^b \delta \phi_\theta \frac{1}{b} GJ \frac{1}{b} \phi_\theta dy \quad (5)$$

$$= \delta \phi_w \underbrace{\frac{4EJ}{b^3}}_{K_w} \phi_w + \delta \phi_\theta \underbrace{\frac{GJ}{b}}_{K_\theta} \phi_\theta \quad (6)$$

$$\delta W_i = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} \quad (7)$$

External Virtual Work

$$\delta W_e = \delta W_{\text{AERO}}^{\text{wing}} + \delta W_{\text{AERO}}^{\text{aileron}} + \delta W_{\text{STRUT}} + \delta W_{\text{INERTIA}} + \delta W_{\text{GRAVITY}}$$

Since we are interested in the divergence dynamic pressure q_D , we can neglect inertia and gravity (static case, no external forces). Thus :

$$\delta W_e = \delta W_A^w + \delta W_A^a + \delta W_{\text{STR}}$$

(1) Wing aerodynamic virtual work δW_A^w The virtual displacement of the AC in z -direction which is kinematically admissible is : $h_{AC}(y) = w(y) + e\theta(y)$; The lift per unit span (perturbation only, so $\alpha_{\text{eff}} = \theta$ and $C_{L_0} = 0$ so $C_L = C_{L/\alpha} \theta$) : $L'_w(y) = q c C_{L/\alpha} \theta(y)$.

$$\delta W_A^w = \int_0^b \delta h_{AC} L'_w(y) dy = \int_0^b \delta(w + e\theta) q c C_{L/\alpha} \theta dy \quad (8)$$

$$= \int_0^b \delta \phi_w \left(\frac{y}{b}\right)^2 q c C_{L/\alpha} \frac{y}{b} \phi_\theta dy + \int_0^b \delta \phi_\theta \frac{y}{b} e q c C_{L/\alpha} \frac{y}{b} \phi_\theta dy \quad (9)$$

Evaluating the integrals :

$$\delta W_A^w = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T q \begin{bmatrix} 0 & \frac{cb}{4} C_{L/\alpha} \\ 0 & \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} \quad (10)$$

(2) Aileron aerodynamic virtual work δW_A^A

The aileron spans from $y = b - b_a$ to $y = b$, with deflection β . The pitching moment coefficient is :

$$C_{m_{AC}} = C_{m_0} + C_{m/\beta} \beta$$

$$\delta W_A^A = \int_{b-b_a}^b \delta h_{AC} L'_{\text{Ail}}(y) dy + \int_{b-b_a}^b \delta \theta M'_{AC_{\text{Ail}}}(y) dy \quad (11)$$

$$= \int_{b-b_a}^b \delta(w + e\theta) q c C_{L/\beta} \beta dy + \int_{b-b_a}^b \delta \theta q c^2 C_{m/\beta} \beta dy \quad (12)$$

Evaluating :

$$\delta W_A^A = \delta \phi_w q e C_{L/\beta} \frac{1}{b^2} \left[\frac{b^3 - (b - b_a)^3}{3} \right] \beta \quad (13)$$

$$+ \delta \phi_\theta q e c C_{L/\beta} \frac{1}{b} \left[\frac{b^2 - (b - b_a)^2}{2} \right] \beta \quad (14)$$

$$+ \delta \phi_\theta q c^2 C_{m/\beta} \frac{1}{b} \left[\frac{b^2 - (b - b_a)^2}{2} \right] \beta \quad (15)$$

In matrix form :

$$\delta W_A^A = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T q \frac{c}{b} \beta \left\{ \begin{array}{c} C_{L/\beta} \cdot \frac{1}{b} \left[\frac{b^3 - (b - b_a)^3}{3} \right] \\ (e C_{L/\beta} + c C_{m/\beta}) \left[\frac{b^2 - (b - b_a)^2}{2} \right] \end{array} \right\} \quad (16)$$

(3) Strut virtual work δW_{STR}

From PCVW applied in the axial direction of the strut :

$$\begin{aligned} F_S &= -\frac{EA}{\ell} u, \quad \ell = \frac{z_A}{\sin \gamma}, \quad u = h_B \sin \gamma \\ F_S &= -\frac{EA}{z_A} \sin^2(\gamma) h_B, \quad F_{Sz} = F_S \sin(\gamma) \\ \boxed{F_{Sz} &= -\frac{EA}{z_A} \sin^3(\gamma) h_B} \end{aligned} \quad (17)$$

Defining the normalized strut spanwise position :

$$\eta_B = \frac{z_A}{b \tan(\gamma)}$$

The displacement at the strut attachment point B :

$$\begin{aligned} h_B(x_B) &= \left(\frac{y_B}{b} \right)^2 \phi_w - x_B \frac{y_B}{b} \phi_\theta, \quad y_B = \frac{z_A}{\tan \gamma} \\ h_B(x_B) &= \frac{z_A}{b \tan \gamma} \left[\frac{z_A}{b \tan \gamma} \phi_w - x_B \phi_\theta \right] = \eta_B [\eta_B \phi_w - x_B \phi_\theta] \\ \delta W_{STR} &= \delta \phi_w K_{STR} \eta_B^4 \phi_w - \delta \phi_w K_{STR} \eta_B^3 x_B \phi_\theta \\ &\quad - \delta \phi_\theta K_{STR} \eta_B^3 x_B \phi_w + \delta \phi_\theta K_{STR} \eta_B^2 x_B^2 \phi_\theta \end{aligned} \quad (18)$$

$$\boxed{\delta W_{STR} = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T (-K_{STR} \eta_B^2) \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}} \quad (20)$$

where $K_{STR} = \frac{EA}{z_A} \sin^3(\gamma)$, $\eta_B = \frac{z_A}{b \tan(\gamma)}$.

Application of PVW : $\delta W_i = \delta W_e$

$$\delta \phi^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \phi = \delta \phi^T \left(q \begin{bmatrix} 0 & \frac{cb}{4} C_{L/\alpha} \\ 0 & \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} + q \frac{c}{b} \beta \mathbf{F}_{Ail} - K_{STR} \eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \right) \phi \quad (21)$$

Full Stiffness Matrix and Equilibrium

The full system (non-forced, static for q_D computation) :

$$\begin{bmatrix} K_w + K_{\text{STR}} \eta_B^4 & -K_{\text{STR}} \eta_B^3 x_B \\ -K_{\text{STR}} \eta_B^3 x_B & K_\theta + K_{\text{STR}} \eta_B^2 x_B^2 \end{bmatrix} - q \begin{bmatrix} 0 & \frac{cb}{4} C_{L/\alpha} \\ 0 & \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} = \mathbf{0}$$

Defining :

$$\mathbf{K}_S = \begin{bmatrix} K_{S_1} & K_{S_{12}} \\ K_{S_{12}} & K_{S_2} \end{bmatrix}$$

with :

$$\begin{aligned} K_{S_1} &= K_w + K_{\text{STR}} \eta_B^4 = \frac{4EJ}{b^3} + \frac{EA}{z_A} \sin^3(\gamma) \eta_B^4 \\ K_{S_{12}} &= -K_{\text{STR}} \eta_B^3 x_B = -\frac{EA}{z_A} \sin^3(\gamma) \eta_B^3 x_B \\ K_{S_2} &= K_\theta + K_{\text{STR}} \eta_B^2 x_B^2 = \frac{GJ}{b} + \frac{EA}{z_A} \sin^3(\gamma) \eta_B^2 x_B^2 \end{aligned}$$

Divergence Dynamic Pressure q_D

For divergence, the system is unforced ($\beta = 0$). The condition is :

$$\det(\mathbf{K}_S - q_D \mathbf{K}_A) = 0$$

where $\mathbf{K}_A = q \begin{bmatrix} 0 & \frac{cb}{4} C_{L/\alpha} \\ 0 & \frac{ecb}{3} C_{L/\alpha} \end{bmatrix}$.

Expanding the determinant :

$$K_{S_1} K_{S_2} - q_D K_{S_1} K_{A_2} - (K_{S_{12}})^2 + q_D K_{S_{12}} K_{A_{12}} = 0$$

$$q_D (K_{S_1} K_{A_2} - K_{S_{12}} K_{A_{12}}) = K_{S_1} K_{S_2} - (K_{S_{12}})^2$$

$$q_D = \frac{K_{S_1} K_{S_2} - (K_{S_{12}})^2}{K_{S_1} K_{A_2} - K_{S_{12}} K_{A_{12}}} = \frac{K_{S_1} K_{S_2} - (K_{S_{12}})^2}{K_{S_1} K_{S_2} - (K_{S_{12}})^2} \cdot \frac{K_{S_1} K_{S_2} - (K_{S_{12}})^2}{K_{S_1} K_{S_2} - K_{S_{12}} K_{A_{12}}} \quad (22)$$

More explicitly :

$$q_D = \frac{K_{S_1} K_{S_2} - (K_{S_{12}})^2}{K_{S_1} K_{A_2} - K_{S_{12}} K_{A_{12}}}$$